

Tucker Vana

Engineering case packet

My work combines technical planning with hands-on setup, operation, and repair. These cases show specific decisions and my contribution.

CASE 01 / Pouch Factory

Production air supply

Air supply selected for initial production, with allowance for a planned second line and a constrained startup budget.

First line supplied

CASE 02 / Fabrication / CNC

Waterjet pump rebuild

An intermittent cutting fault led to pump inspection, a standard scheduled rebuild, and resumed production.

3 working days

CASE 03 / Maybell Quantum

Prototype instrumentation

Instrument readout integration and gas-valve controls shared a GUI for the team's refrigeration prototype.

One operator interface

B.S. Mechanical Engineering, Colorado School of Mines, December 2021.
Cum laude; 3.65 GPA.

Earlier practical work: truck driving, forklift/skid-steer operation, material handling, and yard work at DBC Irrigation Supply.

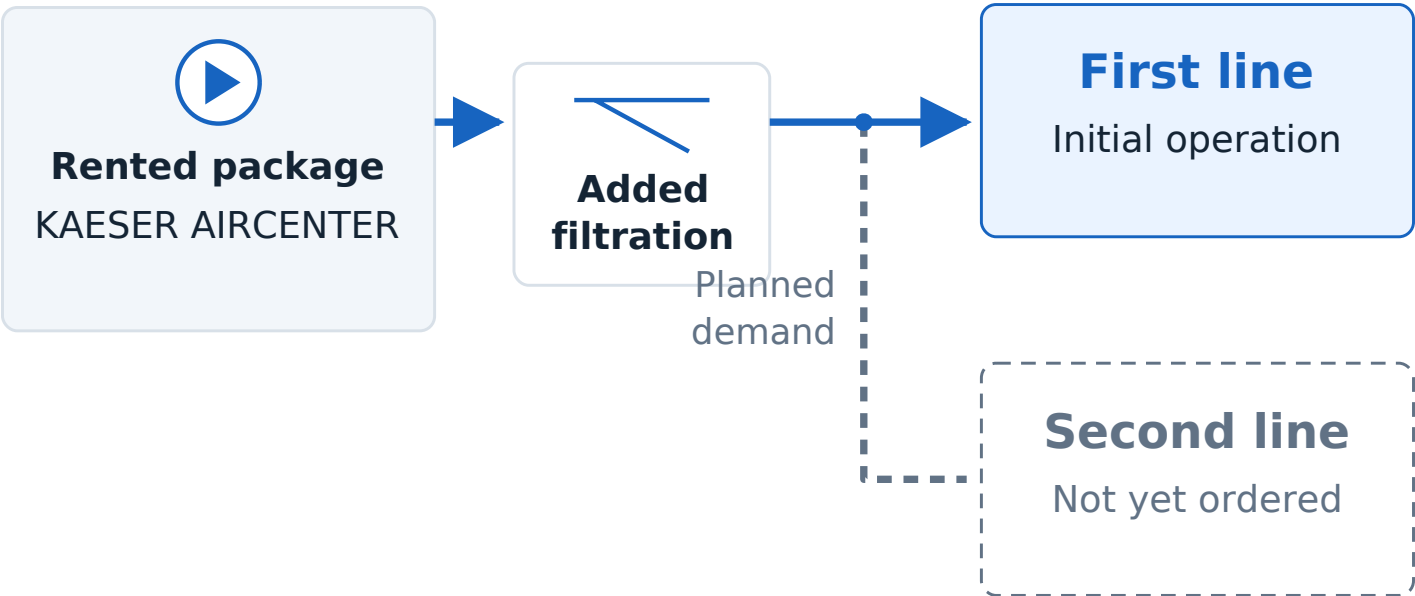
Supply the first line.

Allow for expansion.

I sized the air supply from machine requirements, selected the rented package, and integrated filtration and distribution.

First line supplied
Break-in and initial production run · reported in my account

Functional air-supply arrangement



Solid path: operated first-line load. Dashed path: expansion allowance.

Decisions and contribution

Demand	Peak machine demand + planned second-line allowance + headroom.
Budget	Rented the package for lower initial expenditure than purchase options.
Receiver	No additional receiver purchase; higher compressor duty cycle accepted.

Functional reconstruction from my account. Capacity figures and physical pipe routing are unspecified.

Investigate the cut fault.

Rebuild and restart.

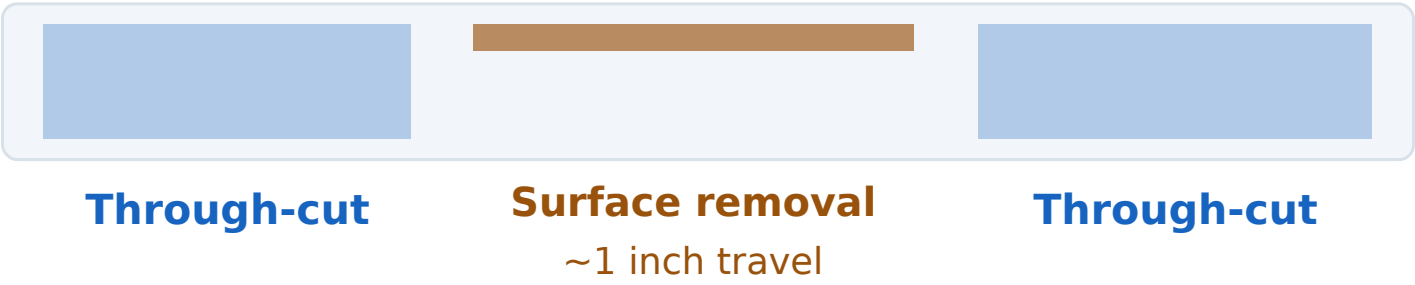
I inspected the high-pressure pump, performed its scheduled rebuild, reassembled it, and checked the machine after repair.

3 working days

Shutdown to production restart · reported in my account

Reported cutting pattern and supply path

Cut travel →



The colored strip sketches penetration; it is not a measured cut-depth trace.

What I found and did

Finding	More mechanical play than expected in the pump cylinder assemblies.
Action	Standard 5,000-hour rebuild kit, reassembly, and machine checks.
Checks	Consistent through-cuts returned; pump output readings were higher.

Qualitative reconstruction. The exact failed component, pump model, and numerical pressure readings are not retained.

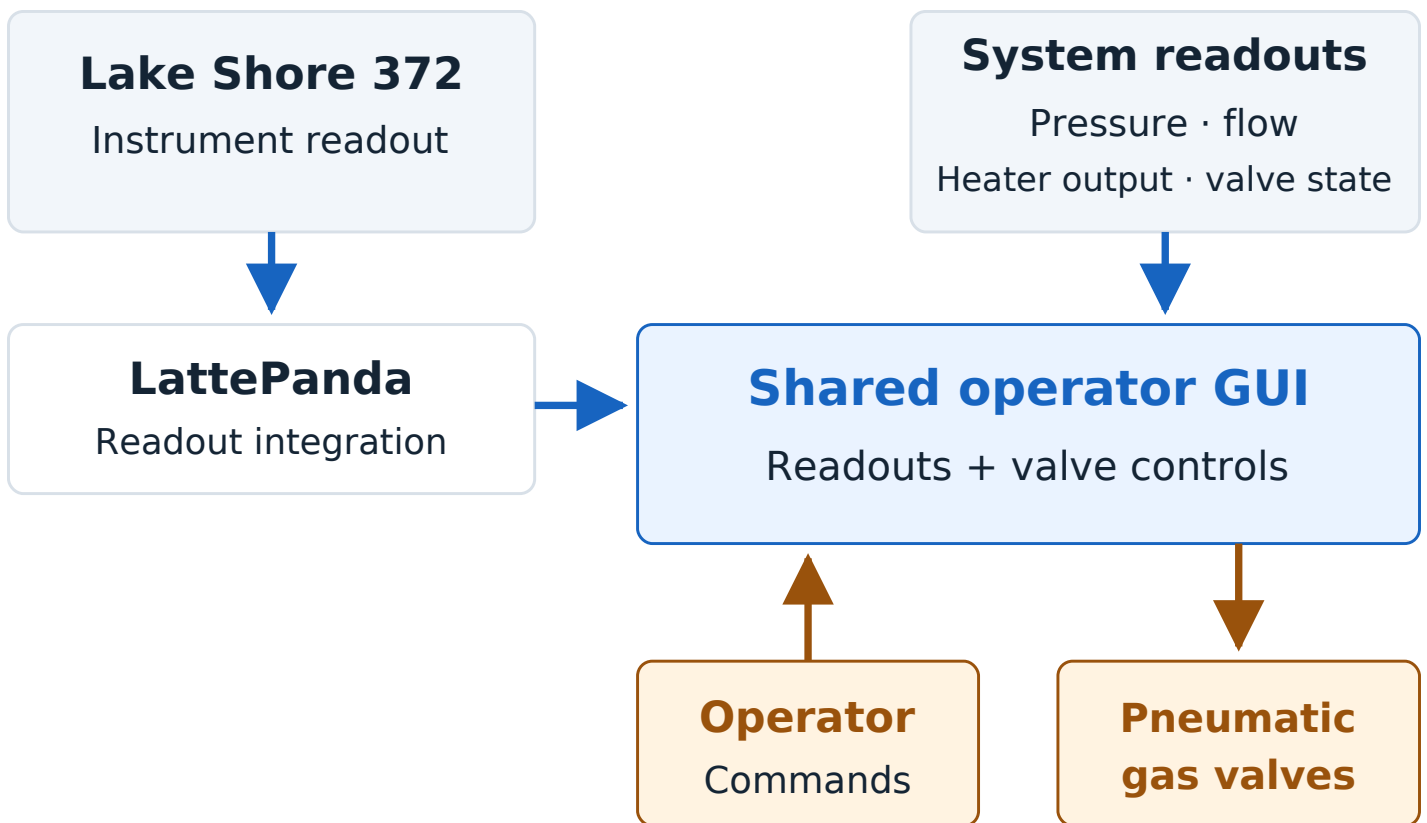
Bring readouts together. Give operators control.

I set up and wired the Lake Shore 372, integrated its readout through a LattePanda, and combined readings and valve controls in one GUI.

One operator interface

System readouts and pneumatic-valve commands · project account

Measurement and operator-command paths



Blue: readouts

Orange: operator commands

Decisions and contribution

My part Instrument setup and wiring, readout integration, and GUI/control connections.

Related work Connector trials; CNC-welder jogging integration. Setup lacked required precision.

Functional reconstruction. Acquisition protocols, valve-driver electronics, timing, and accuracy are unspecified.

Source: [projects/cryogenic-hardware.md](#)